Quantum video classification leveraging textual video representations

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***Abstract*—Quantum video classification is an emerging area of research which unites machine learning and quantum computing principles. This paper describes an approach for quantum video classification which combines video captioning and quantum natural language processing (QNLP) techniques. The proposed approach involves generating captions for videos and using these captions to produce quantum circuits, which are fed as inputs to a classifier running on a quantum simulator. This technique essentially reduces the quantum video classification problem into a quantum text classification problem. The text-based quantum circuits generated for videos also serve the purpose of acting as a new type of dataset for further research on quantum video processing. The proposed quantum video classifier performs well, with an accuracy of 89 percent when trained and tested on a subset of the kinetics dataset.**

Keywords—Quantum Natural Language Processing, Quantum computing, mPLUG-2, lambeq, Quantum Video Classification, qubit, Quantum Video Processing, quantum entanglement, superposition, string diagram.

I. INTRODUCTION

Quantum computing has noticed a massive surge in interest during recent times due to the exponential increase in computational power that it provides over classical computing techniques. This is attributed to two unique aspects of quantum behavior, superposition and entanglement. Qubits, which are the basic information representation units in quantum computers, exhibit superposition by existing in states of 0 and 1 simultaneously, in contrast to classical computers where bits can represent either 0 or 1 at a time [1]. Qubits exhibit entanglement when they are tied together such that a single qubit's state can't be defined without knowing the other qubit's state. This correlation exists even when the entangled qubits are spatially separated [1]. Advances in quantum machine learning and QNLP research have been propelled by the advent of frameworks such as Qiskit [2], Cirq [3] and Lambeq [4], which allow researchers to simulate quantum behavior.

The research area of quantum video classification has been explored by very few researchers at this time, with just two articles available on the topic [5] [6]. This is due to presence of significant challenges which are present when dealing with videos using quantum computing principles. The most significant hurdle is quantum hardware limitations, which arise from the fact that quantum computers currently support a small number of qubits and are susceptible to errors and noise. This current state of quantum computers is classified as the Noisy Intermediate-Scale Quantum (NISQ) era [7]. Encoding videos for classification requires a large number of qubits and lower error rates. The methodology proposed in [6] provides a way to perform video classification with a low qubit count, but the accuracy suffers. Noise and error rates are a common problem faced by all forms of quantum encoded data and not specific to videos.

Further exploration was done to explore the aspects of video classification which require a high qubit count. Video encoding for classification involves representation of pixel values, spatial-temporal information, color values [8]. Complex video features such as object shapes, movements, textures need to be extracted [9]. Video classification requires analyzing the temporal evolution of scenes and objects. The relationships between frames are vital and needs to be encoded and processed [10]. These tasks require a large number of qubits. Moreover, achieving meaningful entanglement patterns between qubits to capture complex correlations within video data require a high qubit count.

The above-mentioned challenges involved in handling videos on quantum computers form the basis for this paper, which introduces the idea of solving the quantum video classification problem by converting it into a quantum text classification problem. With recent advances in video captioning techniques, accurate textual descriptions of videos can be generated. It describes the visual and audio information in a video. QNLP, in contrast to quantum video processing, has witnessed rapid advancement in recent times. This is due to the fact that text data, being lower in dimensionality is easier to encode into a quantum state and utilizes lesser number of qubits. The current NISQ era hardware is sufficient to experiment on many classical NLP problems such as text classification, sentiment analysis etc.

In this paper, quantum video classification is performed on 100 videos from 2 classes using a multi stage pipeline. The first stage involves conversion of videos to text using a state of the-art caption generation model. The second stage involves conversion of these captions to quantum circuits. The third stage involves training and testing a classifier on a quantum simulator with a dataset consisting of quantum circuits from the previous stage.

The remaining sections of the paper are structured as follows: In Chapter 2, we discuss previous studies and current trends in the fields of quantum video classification, video captioning, QNLP and Quantum Machine Learning; The multi-stage pipeline for quantum video classification is described in depth in Chapter 3; Chapter 4 describes the setup used for performing the classification experiment and discusses its results in detail; Chapter 5 concludes the work done and describes future scope and possible enhancements.

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II. Related Work

A. Quantum Video Classification

The first quantum video classification approach involved a quantum 3D Convolutional Neural Network (Q3D-CNN) implementation [5]. The authors introduce an efficient quantum preprocessing step and describe the implementation of a quantum counterpart for a classical Convolutional Neural Network (CNN). The quantum preprocessing step is used to extract information about the differences between successive frames in a video. The quantum CNN is implemented with quantum counterparts for all the elements of a classical CNN including the activation function, kernels, convolution and input matrices. The accuracy of classification by the Q3D-CNN is slightly low in comparison to its classical counterpart. However, the authors demonstrate that the quantum classifier is more efficient compared to its classical counterpart, which gives it an upper hand when it comes to time-sensitive use cases.

The most recent research on quantum video classification introduces the idea of using a quantum process to reduce the dataset before inputting it to a classifier [6]. The authors demonstrate a reduction process which involves extraction of the most essential parts of the video to produce a reduced dataset. The process converts a video which is represented as a matrix with the dimensions (height*width*frames) to a quantum state of size 2^q, where q represents the number of qubits. The authors show the variation in classification results with change in number of qubits. The reduction process also aims to handle the issues involved in data loading, which occurs when transferring massive amounts of data from a classical computer to a quantum computer [11].

B. Video captioning

There are several video captioning benchmark datasets such as Microsoft Research Video Description Corpus (MSVD) [12] and Microsoft Research Video To Text (MSR-VTT) [13]. MSVD consists of 120,000 manually compiled sentences describing 2000 short videos. MSR-VTT contains 10,000 videos from 20 different classes with 20 manually compiled sentences for each video. The quantity of data along with the diversity of video classes including sports, music, cooking etc. makes MSR-VTT the first choice for testing video captioning models.

The Consensus-based Image Description Evaluation (CIDEr) is an evaluation metric which measures how well the generated captions agree with human-generated reference captions [14]. This is used as a key metric for measuring the performance of video captioning models. It evaluates the semantic relevance of the generated captions by referring to how well the captions capture the main objects, actions, and relationships present in the video. It also assesses the similarity between the n-grams (sequences of n words) in the generated captions and those in the reference captions. CIDEr uses term frequency (TF) normalization to ensure that frequent words do not dominate the evaluation and that a wider range of vocabulary is considered when assessing caption quality. The following paragraph summarizes video captioning models which have the highest CIDEr score on MSR-VTT.

The Vision Audio Language Omni Perception pretraining model (VALOR) introduced a novel architecture for performing video captioning and other multimodal tasks by using an end-to-end modelling of vision, text and audio [15]. The architecture describes three standalone encoders for vision, text and audio, a decoder for text generation. It achieves a CIDEr score of 74.0 for video captioning task when tested using the MSR-VTT dataset. Video Language pre-training by feature Adapting and Blending (VLAB) is another architecture which provides good results for video captioning when tested using the MSR-VTT dataset, with a CIDEr score of 74.9 [16]. It addresses the limitations of Contrastive Language Image Pre-training (CLIP) representations [17]. Vision-Audio-Subtitle-Text (VAST) model provides a superior video captioning method by training audio and visual captioning models separately on a custom dataset of 27 million clips (VAST-27M) [18]. The generated captions and subtitles are combined using a large language model. A CIDEr score of 78.0 on MSR-VTT is achieved by this video captioning approach. mPLUG-2 introduces a modularized approach to solving problems involving text, images and videos [19]. The flexibility offered by this approach to select and combine the best modules leads to this model achieving state of the art results on a variety of tasks including video captioning. The CIDEr score it obtains on the MSR-VTT dataset is 80.3.

C. QNLP and Quantum Machine Learning

The DisCoCat framework provides a way to relate linguistic features and quantum states. This is achieved by representing sentences as string diagrams. It leverages Lambek's pregroup grammar [20] to represent sentence level meaning. The first practical QNLP experiment [21] was performed in 2020, which involves a quantum sentence classifier. Training for this model was carried out using dataset consisting of 16 sentences. The experiment was scaled by increasing the size of the training dataset to 130 sentences [22].

There are several toolkits available for quantum machine learning. TensorFlow Quantum [23] is one such framework which combines the capabilities of Cirq [3] and TensorFlow. Quantum circuits can be designed and implemented with the help of Cirq, while classical machine learning can be accomplished with TensorFlow. TensorFlow Quantum basically extends TensorFlow's functionality by adding support for workload to be run on Quantum Processing Units (QPUs).

Pennylane [24] is another python library which facilitates development of Quantum Machine Learning algorithms. The main feature that it provides is computation of gradients for variational quantum circuits. This allows for training of quantum machine learning models using classical optimization techniques. Variational quantum circuits leverage both classical and quantum computing power. They can undergo objective-specific optimization by classical computers.

Lambeq [4] is the first QNLP library which provides a pipeline for conversion of sentences to quantum circuits. While PennyLane and TensorFlow Quantum can be used for all Quantum Machine Learning applications, Lambeq is specific to QNLP. The pipeline that lambeq provides is very similar to the process followed in [21] and [22]. Lambeq is modular and can be customized with different options for each stage of the pipeline. It also offers optimization options to reduce the number of qubits required by rewriting string diagrams.

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III. METHODOLOGY

The paper proposes a three-stage pipeline for Quantum Video Classification. The three stages are video captioning, quantum circuit generation and quantum classification. Stages one and two are performed on classical hardware and stage three is performed on a quantum simulator running on classical hardware. Fig. 1 depicts the different stages of the pipeline with the inputs and outputs for each stage. The model used for video captioning in stage one is mPLUG-2 and lambeq is the framework used in stages two and three for QNLP tasks. Each of the three subsections following this paragraph describe a pipeline stage in detail.

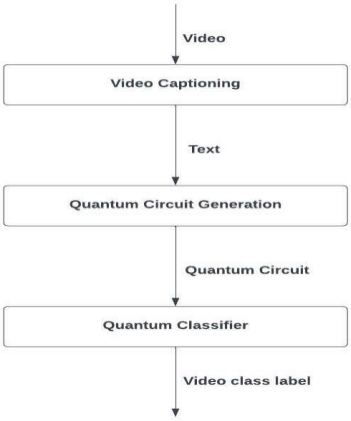


Fig. 1. Pipeline for Quantum Video Classification

A. Video Captioning

The preference of mPLUG-2 over other video captioning models in this stage can be attributed to the fact that it currently achieves the highest CIDEr score when tested on MSR-VTT dataset. There are multiple pre-trained mPLUG-2 models available for different tasks. The mPLUG-2 model, which is pre-trained on the WebVid-2M dataset with 2.5M video-text pairs for video captioning task is used in this paper. The modules and flow of data in mPLUG-2 varies based on the task it is trained for. Fig. 2 depicts the modules used in mPLUG-2 for video captioning and a sample of inputs and outputs for this stage. Inputs to the pretrained model include the video for which captions need to be generated, and a sentence such as "generate captions for this video" or "generate a description for this video".

The video encoder module employs temporal locality to decompose the visual representation into its spatial and temporal components. This is done to overcome the problem of modelling spatial-temporal relationships without a lot of pre-training. The architecture of this module is split into a transformer block for spatial modelling and a submodule for temporal modelling. The temporal modelling is done by using a technique called multi group fusion, which enables the learning of detailed temporal insights by processing distinct representation subspaces at various points in time.

The text encoder module uses Bidirectional Encoder Representations from Transformers (BERT). BERT makes use of the transformer architecture along with bidirectional training mechanism which uses Masked Language Modelling (MLM) to randomly mask words in a sentence and predict it using the words present before and after it. This simultaneous consideration of left and right neighbors of a word leads to better contextualized representations.

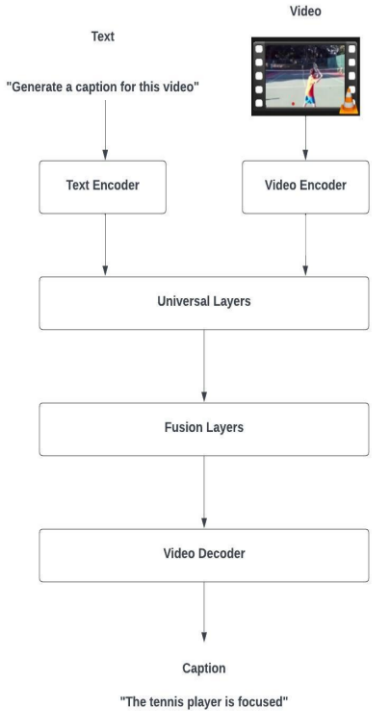


Fig. 2. Video captioning modules in mPLUG-2

Another training strategy used is Next Sentence Prediction where two statements are given as input and the model tries to predict if the second statement comes immediately after the first in the real context. This leads to the BERT learning relationships at a sentence level.

Textual and visual representations are modelled in the same semantic space using the universal layers, without losing their initial representations. The universal layer is a novel approach introduced by mPLUG-2. The fusion layers use transformer blocks to produce multimodal representations of text and vision. The transformer uses cross-attention layers to align visual and textual tokens which help the encoder understand the relationship between text and vision better. The final module consisting of a multipurpose decoder offers great flexibility for pretraining the model for basic unimodal tasks like classification as well as more complex cross-modal generative applications including video captioning and visual question answering. It houses a transformer architecture and acts as a video decoder when used for the task of video captioning. It outputs one or more sentences describing the video. The output sentence length and the total number of sentences are adjustable.

B. Quantum Circuit Generation

The use of lambeq to transform a caption into a quantum circuit involves three steps which are described in this section. The caption is first transformed to a string diagram, which is rewritten to produce a computationally efficient version, and a quantum circuit is obtained from this rewritten string diagram. Lambeq's string diagrams effectively represent the meaning of sentences, which is derived from the relationship between its constituent words. This relationship is defined using Lambek's pre-group grammar rules. The string diagram for the caption, "The tennis player is focused" is shown in Fig. 3. Boxes represent words, straight lines represent wires and curves represent cups.

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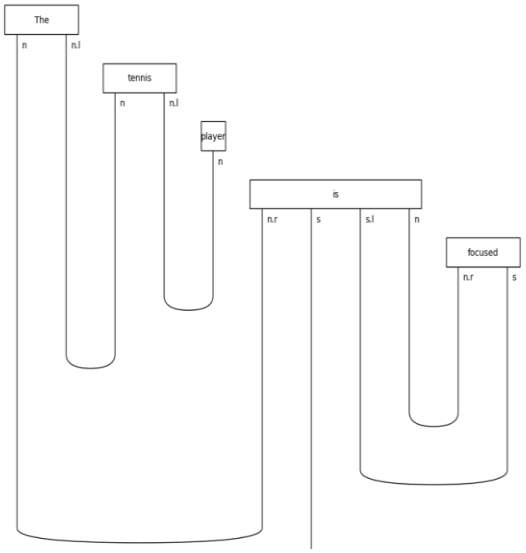


Fig. 3. String diagram for the caption “The tennis player is focused"

According to the Lambek pre-group grammar, the wires are labelled with types or adjoints that illustrate the relationships between the words in a statement. The types used here are “*n*” which stands for noun and “*s*” which stands for sentence. A left and right adjoint exist for each type. The type ‘n’ has adjoints “ n^l ” and “ n^r ”, the type “*s* ” has adjoints “ s^l ” and “ s^r ”. The string diagram can be explained using the rules shown “(1)” and “(2)” which are defined for any type “t”. The syntactic derivation for the string diagram in Fig. 3 is shown in “(3)” using these 2 rules. The result “s” indicates the grammatical correctness of the sentence according to the pre-group rules. Bobcat parser and DisCoPy are used in lambeq to handle syntactic derivations and string diagrams.

$$t^l \,.\, t \rightarrow 1 \rightarrow t \,.\, t^l \tag{1}$$

$$t \,.\, t^r \rightarrow 1 \rightarrow t^r \,.\, t \tag{2}$$

$$n \,.\, n^l \,.\, n \,.n^l \,.\, n \,.\, n^r \,.\, s \,.\, s^l \,.\, n \,.\, n^r \,.\, s \rightarrow \mathrm{s} \tag{3}$$

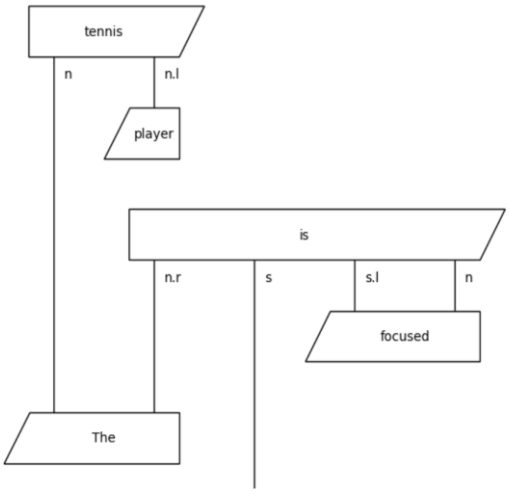


Fig. 4. Rewritten String diagram for the caption “The tennis player is focused"

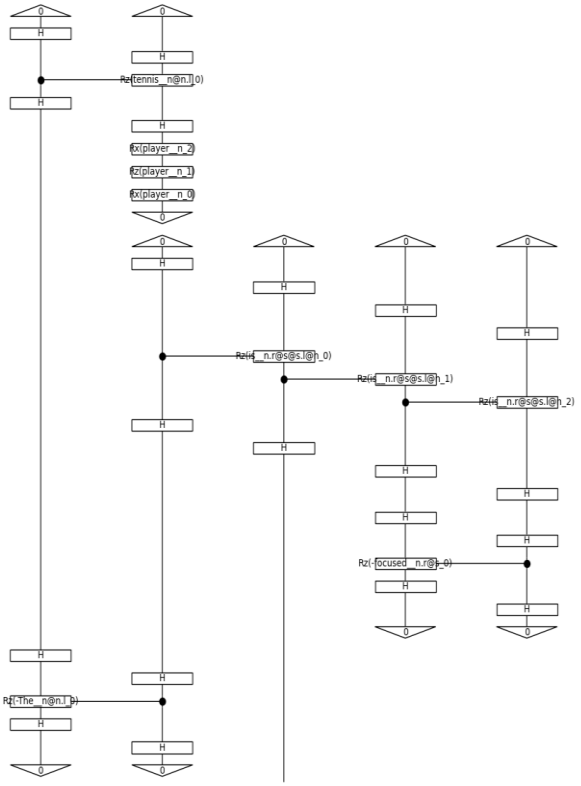


Fig. 5. Quantum Circuit for the caption “The tennis player is focused"

The string diagram in Fig. 3 is further optimized to reduce the amount of computation required. Lambeq has a set of rewriting rules that it uses to reduce the number of syntactic derivations required. Fig. 4 shows the rewritten string diagram. One of the outcomes of rewriting is the removal of cups which which significantly reduces the number of qubits required. Each cup removed reduces the qubit count by 1. Fig. 5 shows the quantum circuit which is generated from the rewritten string diagram. ‘H’ represents Hadamard gates, the triangles on top represent states and the traingles at the bottom represent effects. The generation of quantum circuits is done with the help of ansatz, a map in lambeq which determines the number of qubits for wires and types. Ansatz also determines the parameters for quantum states corresponding to each word. The quantum circuit shown in Fig. 5 requires 6 qubits.

C. Quantum Classifier

Quantum circuits for all videos in the dataset are generated and class labels for these circuits are the same as their source videos. The circuits and their class labels constitute the dataset to perform classification. The dataset is proportioned into sets for training, testing and validation, which are input to the classifier. The NumPy model from lambeq is used along with a JAX backend for simulating quantum classification. A binary classifier is implemented and the optimizer used is Simultaneous Perturbation Stochastic Approximation (SPSA). This calculates new parameters for quantum gates during training. The just-in-time compilation provided by JAX is used by prediction functions.

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IV. Experimental Setup and Results

The kinetics-700 dataset [25] was used to train and test the quantum classifier. The dataset is a benchmark for video classification, containing a variety of human action clips. 100 videos from 2 of the classes in the dataset, “playing tennis” and “playing guitar” were used for this experiment. The length of each clip is 10 seconds. 3 captions were generated using mPLUG-2 for each of the 100 videos. The length of each caption is 5 words. Quantum circuits were generated for a total of 300 sentences equally divided between 2 classes. The quantum circuits were divided into proportions of 70:15:15 for training, testing and validation. The train set contained 210 quantum circuits, the test and validation sets had 45 quantum circuits each.

The video captioning model as well as the quantum simulation was run on a Google Colab with an Nvidia V100 GPU and 51 GB RAM. Peak memory usage of around 30 GB was noticed during quantum simulation. This is because the NumPy model with JAX stores the pre-compiled functions for each circuit in memory. Training the model was done over 3000 epochs and batches were of size 50. The loss and accuracy graphs for train and validation sets are illustrated in Fig. 6. An accuracy of 89 percent was achieved on the test set. TABLE I showcases the experimental results of the model described in this paper and binary classifier variants of other currently available quantum video classification methods [5] [6]. The proposed model, which needed 6 qubits only, provides an improvement over the data reduction model [6], which achieves 59% accuracy with 6 qubits and 73% accuracy with 16 qubits. The comparison of accuracy numbers has to be interpreted with due consideration to the differences in simulation environments, data volume and data domains. The experiment proves that the procedure outlined in this paper provides an effective way of representing videos for quantum processing.

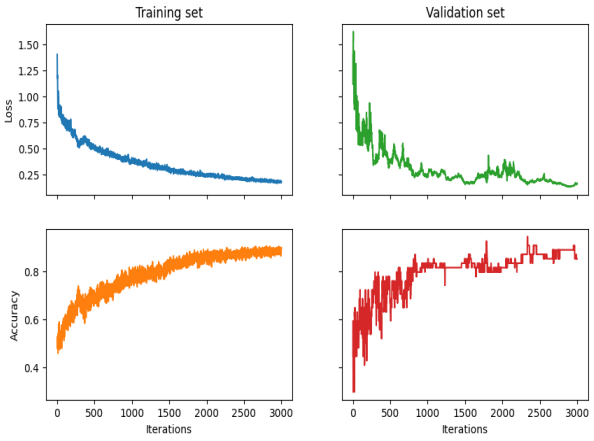


Fig. 6. Loss and Accuracy graphs for training and validation sets

TABLE I. Quantum Video Classification Models Performance

Method	Number of videos used from dataset	Accuracy
QCNN [5]	500	73%
Q3D-CNN [5]	500	83%
Quantum Data Reduction with 16 qubits [6]	240	74%
Quantum Data Reduction with 6 qubits [6]	240	59%
Proposed model with 6 qubits	100	89%

V. Conclusion

The paper provides a novel approach for quantum video classification by reducing the problem into a simpler QNLP task. The simulated results prove that the proposed pipeline is effective and provides a base for further exploration on cross-modal approaches to solving problems involving videos and quantum computers. Efficiently representing visual data on quantum computers is key to advancing research in the field. Further experimentation using the proposed pipeline by including more classes of videos, different datasets and multiple simulation environments would help in gaining more data points for building a state of the art model for handling video related tasks on quantum computers. The work can also be extended to perform multimedia search and retrieval on quantum hardware. Further exploration on parsers and extending rules for rewriting string diagrams could help represent data more efficiently and advance QNLP.

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